A Sensor-Based AIoT System for Real-Time Photography Condition Classification and Editing Guidance

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Abstract*—**This paper presents a lightweight Artificial Intelligence of Things (AIoT) system designed to assist photographers by identifying capture conditions and providing editing guidance. The system utilizes an Arduino Nano 33 BLE Sense Rev2 to collect real-time sensor data, including ambient light and motion measurements. A neural network model is trained to classify four photography conditions: bright-stable, bright-shaky, lowlight-stable, and lowlight-shaky. The trained model is deployed on the microcontroller using TinyML techniques, enabling real-time inference on a resource-constrained device. Based on the predicted condition, the system provides targeted image editing recommendations. Experimental results show that the model achieves 99.05% classification accuracy with efficient real-time performance. The proposed approach demonstrates a practical alternative to traditional image-based processing by leveraging sensor data for photography assistance. **(Abstract***)**

I. Introduction

Photography quality is significantly influenced by environmental conditions such as lighting and camera stability. Poor lighting can lead to underexposed or noisy images, while camera motion often results in blur. These issues typically require manual correction during post-processing, which can be time-consuming and inconsistent.

Recent advancements in artificial intelligence have introduced automated image enhancement techniques. However, most existing methods rely on analyzing image pixels directly, which requires high computational power and is not suitable for embedded systems or real-time use.

This paper proposes a novel sensor-based approach that focuses on analyzing capture conditions instead of image content. By leveraging onboard sensors, the system predicts the environmental condition at the moment of capture and maps it to appropriate editing recommendations. This reduces computational complexity while providing practical assistance for photographers.

II. Related Work

Image enhancement has traditionally relied on computer vision techniques such as histogram equalization, edge detection, and filtering. More recently, deep learning methods, particularly convolutional neural networks (CNNs), have been widely used for tasks such as denoising, super-resolution, and automatic photo enhancement.

Despite their effectiveness, these approaches require significant computational resources and are not suitable for microcontroller-based systems. TinyML has emerged as a solution for deploying machine learning models on embedded devices by focusing on low-dimensional sensor data.

Prior research has also shown that environmental factors such as lighting conditions and camera motion strongly influence image quality. This motivates the use of sensor-based approaches to estimate image conditions without directly processing images.

The proposed system combines TinyML with photography-specific knowledge to create a practical and efficient solution for real-time editing guidance.

III. System Overview

The proposed system implements a complete end-to-end AIoT pipeline that operates in real time on an embedded device.

***System Pipeline*:** Sensor Data → Feature Extraction → Neural Network Model → Condition Prediction → Editing Recommendation

The system consists of the following components:

1. *Sensor Data Acquisition*
   Real-time data is collected from the ambient light sensor and IMU.
2. *Feature Processing*
   Sensor readings are formatted into input features for the model.
3. *Model Inference*
   A trained neural network performs classification directly on the device.
4. *Condition Classification*
   The system predicts one of four photography conditions.
5. *Editing Guidance*
   The predicted condition is mapped to a predefined editing workflow.

This architecture enables efficient real-time operation without requiring image processing on the microcontroller.

IV. Methodology

A. Data Collection

Sensor data was collected using the Arduino Nano 33 BLE Sense Rev2 under controlled conditions representing four scenarios:

- brightstable
- brightshaky
- lowlightstable
- lowlightshaky

Each sample includes:

- Ambient light intensity (light)
- Accelerometer readings (ax, ay, az)
- Gyroscope readings (gx, gy, gz)

The dataset was balanced across all classes to ensure fair training.

B. Feature Processing

Raw sensor values were directly used as model inputs. This approach minimizes computational overhead and supports real-time inference on embedded hardware.

C. Model Design

A lightweight feedforward neural network was implemented using PyTorch:

- Input layer: 7 features
- Hidden layers: 32 → 16 neurons
- Output layer: 4 classes

ReLU activation functions were used, and the model was trained using cross-entropy loss with the Adam optimizer.

D. Edge Deployment

The trained model was converted and deployed onto the Arduino device using TinyML techniques. The deployment process included:

- Model export (PyTorch → ONNX / TFLite)
- Model optimization for embedded execution
- Integration into C/C++ Arduino code
- Real-time inference using sensor input

This enables the system to perform predictions directly on-device without external computation.

V. Experminents and Results

A. Experimental Setup

The system was evaluated in real-world conditions using:

- Hardware: Arduino Nano 33 BLE Sense Rev2
- Sensors: IMU and ambient light sensor
- Inference: real-time on-device execution

The dataset was split using an 80/20 training and testing ratio.

B. Demo Scenario

The system operates in real time as follows:

1. The user holds the device during a simulated photo capture
2. Sensor data is collected instantly
3. The model predicts the capture condition
4. The prediction is displayed via Serial Monitor
5. The user applies the corresponding editing preset

This demonstrates a complete AIoT pipeline from sensing to practical application.

C. Performance Results

The model achieved the following performance:

- Test Accuracy: **99.05%**
- High precision and recall across all classes
- Minimal misclassification observed

The confusion matrix confirms strong separability between all four classes.

D. Latency and Efficiency

- Inference latency: near real-time (milliseconds)
- Memory usage: within microcontroller constraints
- Model size: optimized for embedded deployment

These results demonstrate that the system is suitable for real-time embedded applications.

E. Baseline Comparison

A baseline approach was tested using separate labels for lighting and motion independently.

- Baseline Accuracy: ~75%
- Proposed Method Accuracy: **99%**

Combining lighting and motion into unified classes significantly improved classification performance.

VI. Discussion

The results demonstrate that sensor-based data is highly effective for classifying photography conditions. The improved labeling strategy plays a key role in achieving high accuracy.

A major advantage of this approach is its efficiency. Unlike image-based methods, the system avoids heavy computation and operates directly on low-dimensional sensor data.

However, the system does not analyze image content, which may limit performance in complex scenarios where lighting and motion alone are insufficient indicators of image quality.

VII. PRACTICAL APPLICATION IN IMAGE EDITING

The system provides direct value by guiding post-processing decisions.

Each predicted class corresponds to an editing workflow:

- brightstable → contrast and highlight adjustments
- brightshaky → sharpening and clarity enhancement
- lowlightstable → exposure increase and noise reduction
- lowlightshaky → exposure increase, stronger noise reduction, and sharpening

This reduces manual effort and improves editing consistency.

VIII. CONCLUSION

This paper presented a real-time AIoT system for photography condition classification using sensor data and TinyML deployment.

The system achieves high accuracy while operating efficiently on a microcontroller. By linking sensor-based predictions to editing workflows, it provides a practical solution for photography enhancement.

Future work includes integrating image-based analysis, automating editing pipelines, and expanding the system to support additional conditions.